

Homework: Calculate the Area of a Rectangle Using a Function

Write a C++ program that creates a function named `calculateArea()`.

The function should receive the length and width of a rectangle and return the area.

Formula : $\text{Area} = \text{Length} \times \text{Width}$ $\text{Area} = \text{Length} \times \text{Width}$

Example Input

Enter length : 10

Enter width : 5

Example Output

Area of the rectangle = 50

Requirements

1. Ask the user to enter the length and width.
2. Create a function named `calculateArea()`.
3. Pass length and width to the function.
4. Calculate the area inside the function.
5. Return the area.
6. Call the function from `main()`.
7. Display the result.

```
#include <iostream>
```

```
#include <cmath>
```

```
using namespace std;
```

```
double calculateArea(double length, double width)
```

```
{
```

```
    double area = length * width;
```

```
    return area;
```

```
}
```

```
int main()
```

```
{
```

```
    double length;
```

```
    double width;
```

```
    cout << "Enter length: ";
```

```
    cin >> length;
```

```
    cout << "Enter width: ";
```

```
    cin >> width;
```

```
    cout << "Area of the triangle = " << calculateArea(length, width) << endl;
```

```
    return 0;
```

```
}
```